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EC 551 – Advanced Digital Design with Verilog and FPGAs

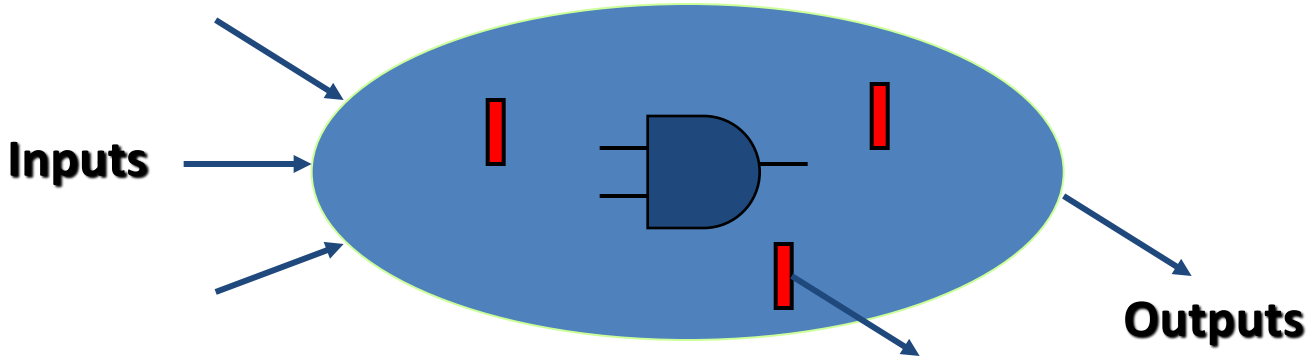
Lecture 12
Sequential Synthesis: Retiming

Prof. Douglas Densmore

Additional material courtesy of: ECE 667 - Synthesis & Verification (school ?) and ELEC 7770 Advanced VLSI Design (Vishwani D. Agrawal; Auburn)

Optimizing Sequential Circuits by Retiming Gate-level Netlist

Netlist of gates and registers:



Various Goals:

- Reduce clock cycle time
- Reduce area
  - Reduce number of latches (registers)

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Retiming

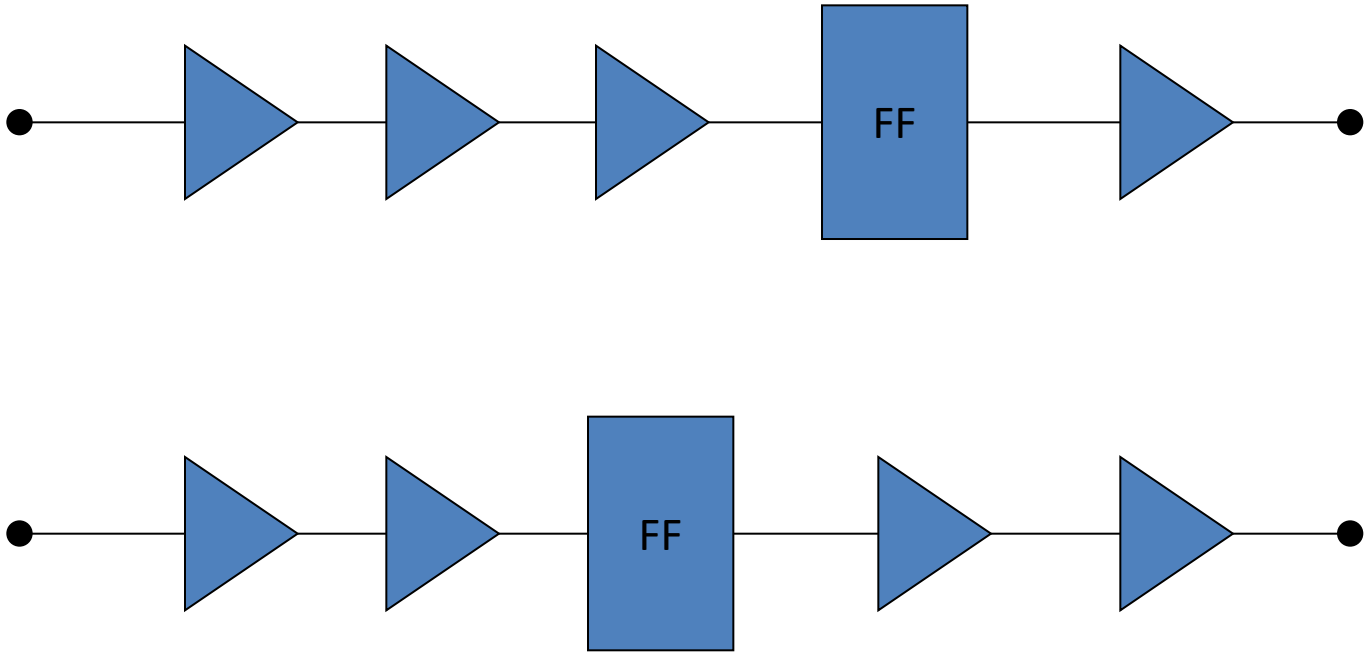
Problem

- Pure combinational optimization can be myopic since relations across register boundaries are disregarded

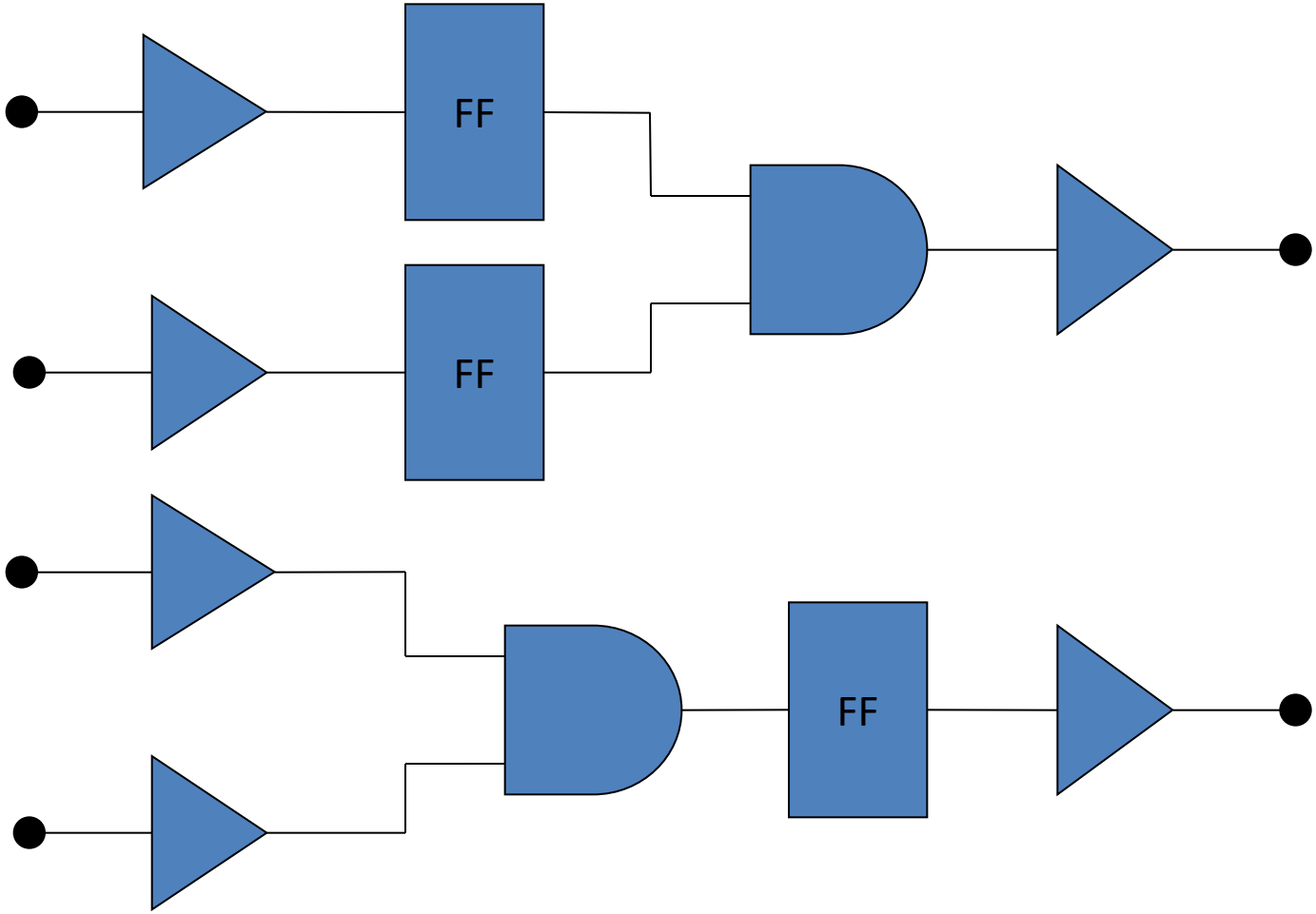
Solutions

- Retiming: Move register(s) so that
  - clock cycle decreases, or number of registers decreases and
  - input-output behavior is preserved
- Peripheral retiming: Combine retiming with combinational optimization techniques
  - Move latches out of the way temporarily
  - optimize larger blocks of combinational

Example: Faster Clock



Example: Reduced Flip-Flops



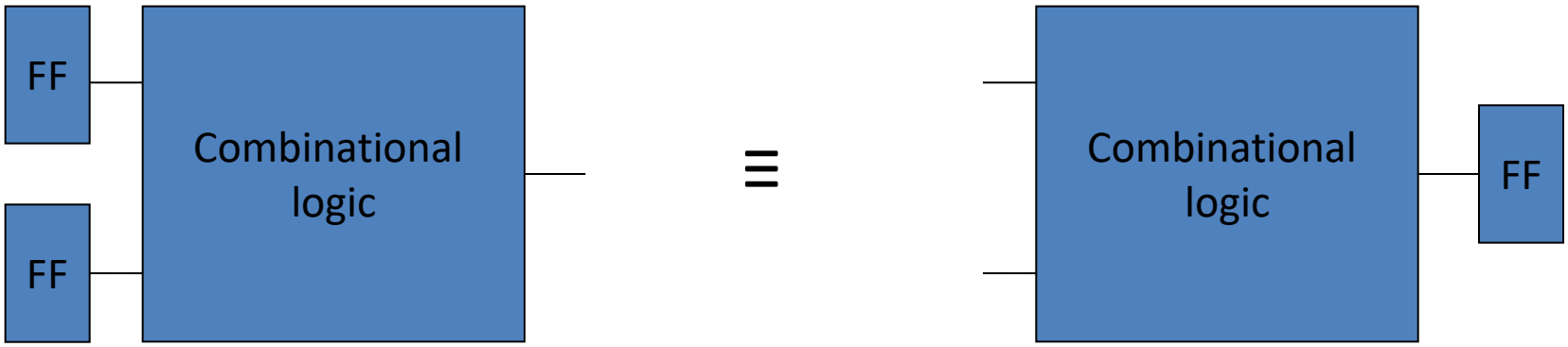
Applications of Retiming

- Performance optimization
- Area optimization
- Power optimization
- Testability enhancement
- FPGA optimization

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Fundamental Operation of Retiming

- A retiming move in a circuit is caused by moving all of the memory elements at the input of a combinational block to all of its outputs, or vice-versa.



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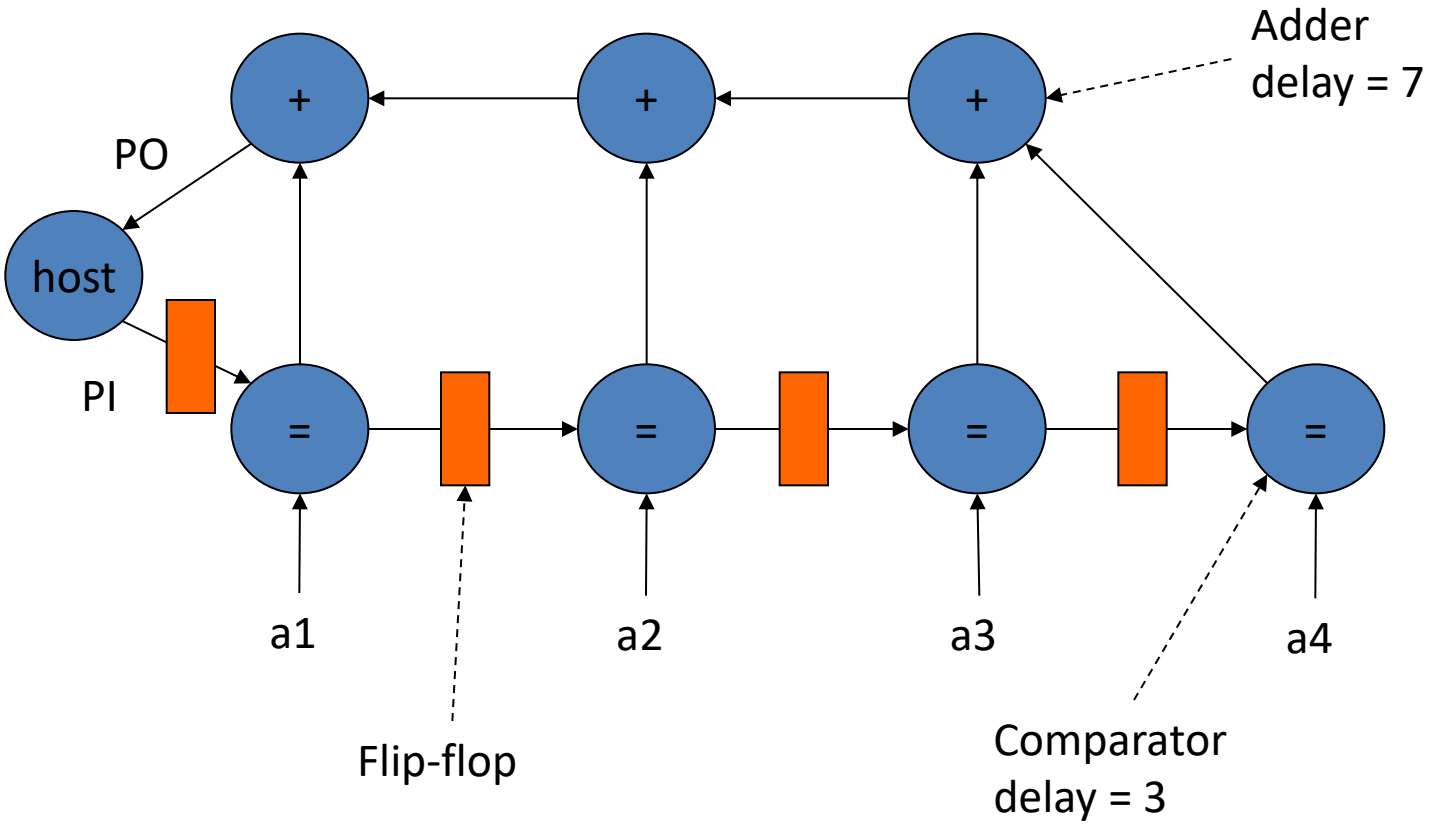
Circuit Representation

[Leiserson, Rose and Saxe (1983)]

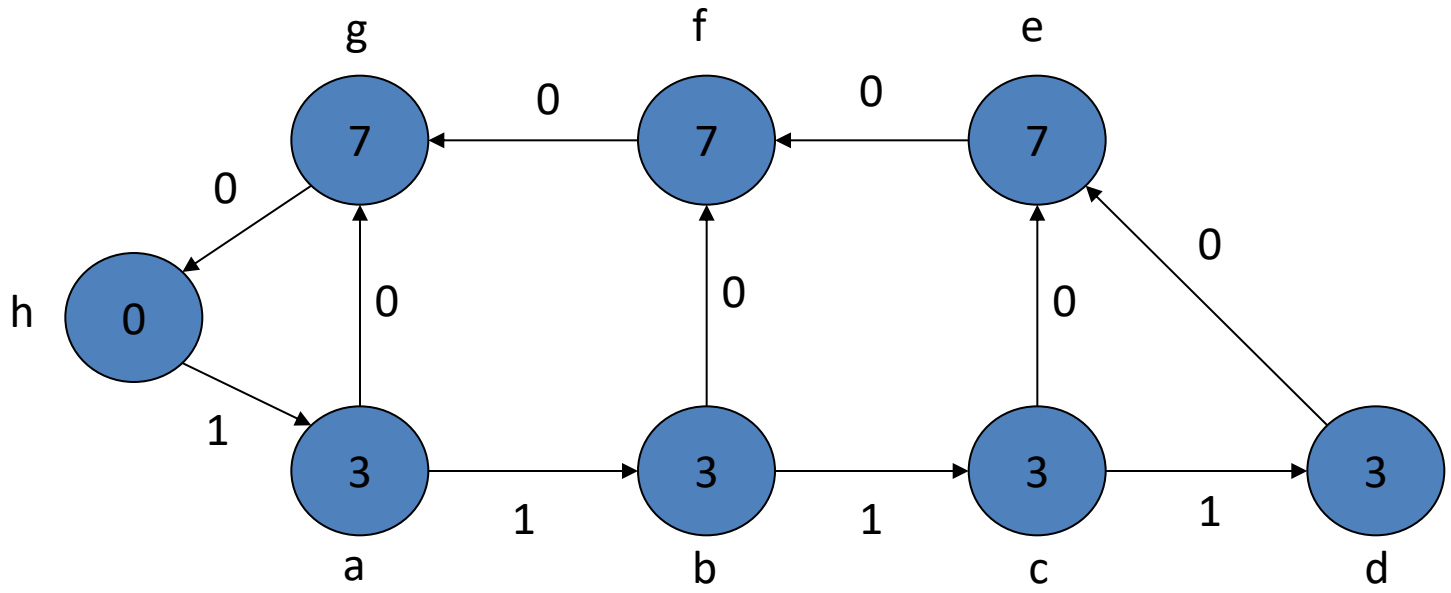
Circuit representation: G(V,E,d,w)

- V $\leftrightarrow$ set of gates
- E $\leftrightarrow$ set of wires
- d(v) = delay of gate/vertex v, $(d(v) \geq 0)$
- w(e) = number of registers on edge e, $(w(e) \geq 0)$

A Correlator Circuit



Graph Model



Vertex, vi, combinational, delay = d(vi), assumed unchanged by retiming

d(host) = 0

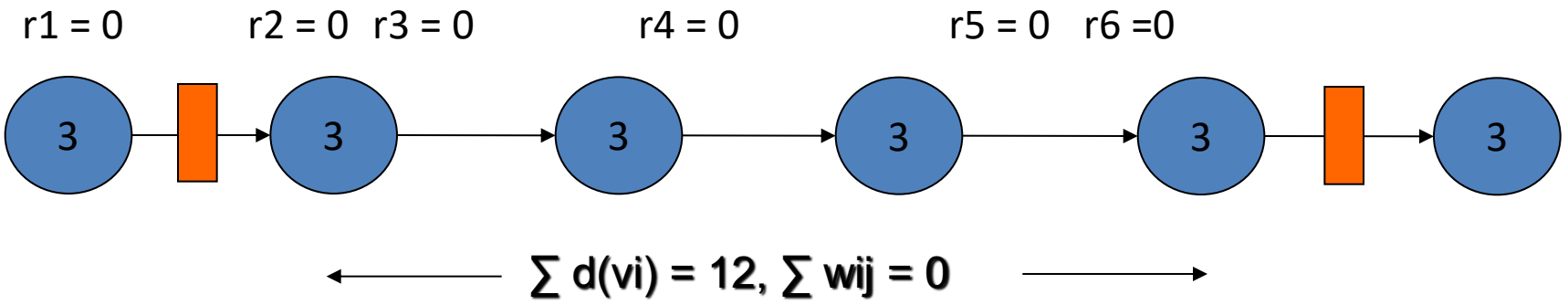
Edge, e(vi,vj) or eij, weight wij = number of flip-flops between vi and vj

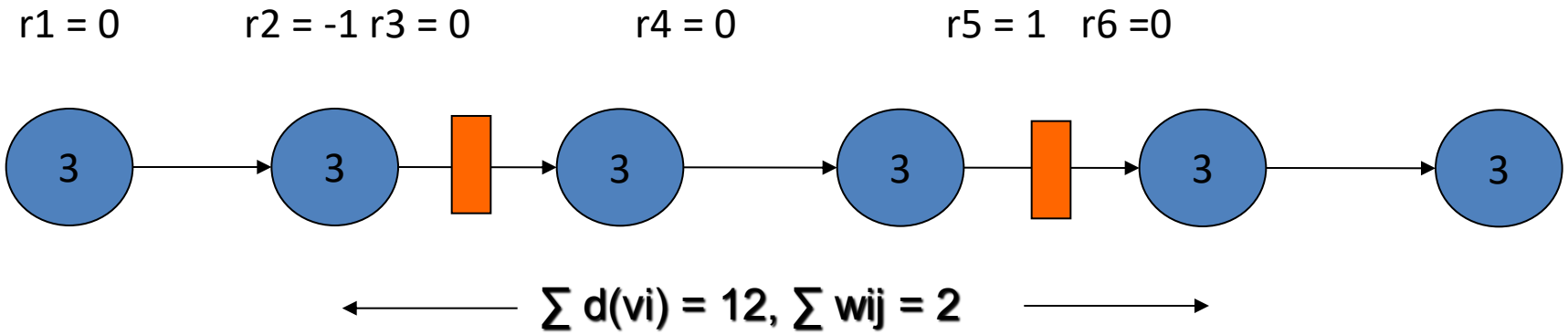
Path Delay and Path Weight

- A set of connected nodes specify a path. A path does not traverse through the host node.
- Path delay = $\sum d(v_i)$ = combinational delay of path
- Path weight = $\sum w_{ij}$ = clock delay of path
- Retiming of a node i is denoted by an integer ri
  - It represents the number of registers moved across, initially ri = 0
  - Register moved from output to input, ri → ri + 1
  - Register moved from input to output, ri → ri – 1
  - After retiming, edge weight wij' = wij + rj – ri

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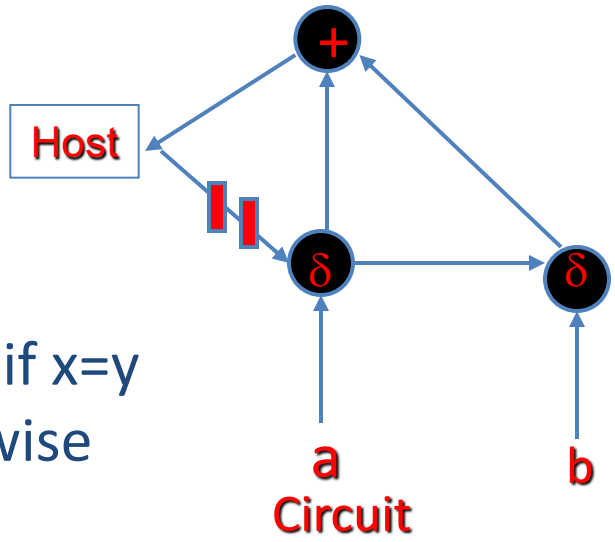
Example of Node Retiming



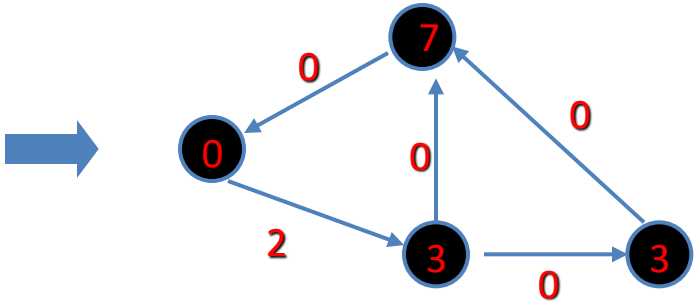


Circuit Representation

Example: Correlator



$\delta(x, y) = 1$ if $x=y$
0 otherwise



Graph (Directed)

Operation	delay
δ	3
+	7

Every cycle in Graph has at least one register i.e. no combinational loops.

Preliminaries

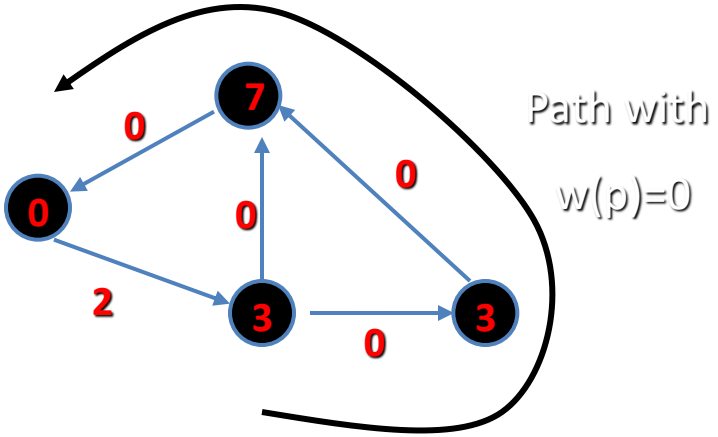
For a path p : $v_0 \xrightarrow{e_0} v_1 \xrightarrow{e_1} \cdots v_{k-1} \xrightarrow{e_{k-1}} v_k$

$$d(p) = \sum_{i=0}^{k} d(v_i)$$ (includes endpoints)

$$w(p) = \sum_{i=0}^{k-1} w(e_i)$$

Clock cycle

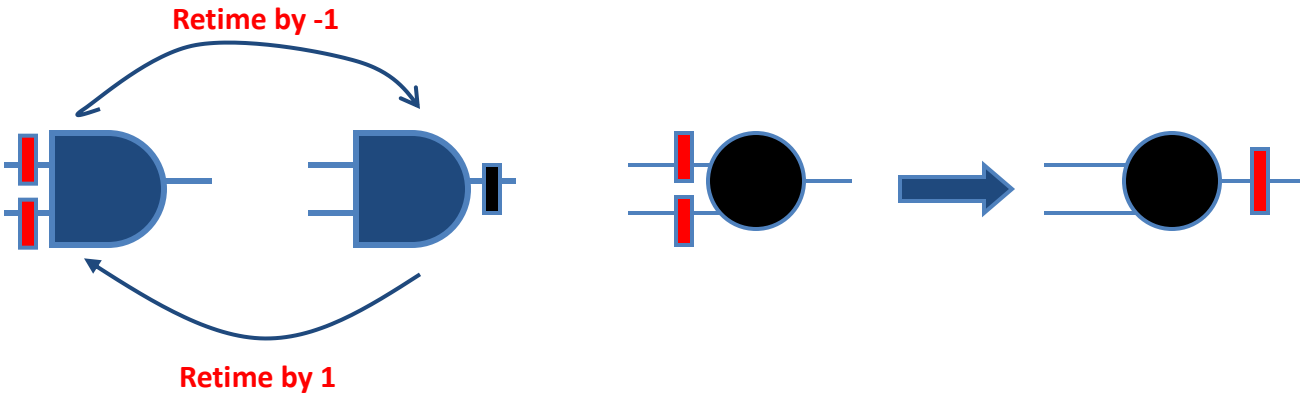
$$c = \max_{p:\, w(p)=0} \{d(p)\}$$



For the correlator circuit: c = 13

Basic Operation

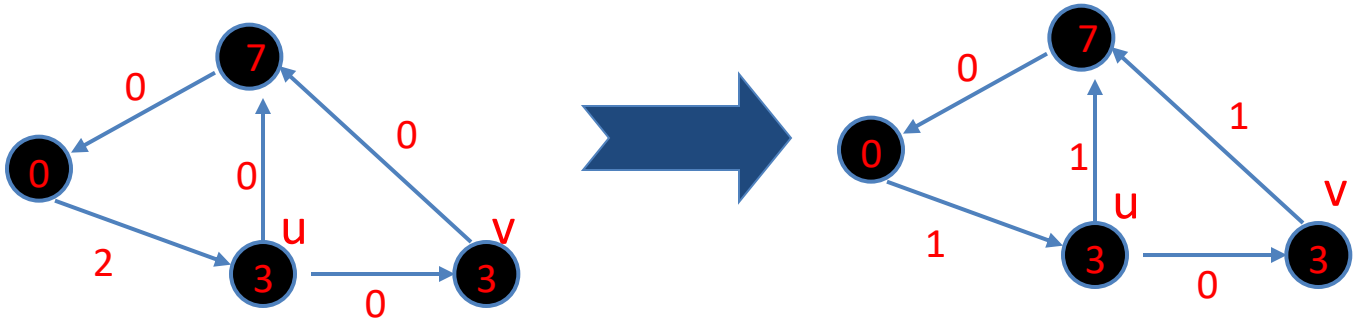
- Movement of registers from input to output of a gate or vice versa



- Does not affect gate functionalities
- A mathematical definition: retardation
  - r: V $\rightarrow$ Z, an integer vertex labeling
  - $w_r(e) = w(e) + r(v) - r(u)$ for edge $e = (u,v)$

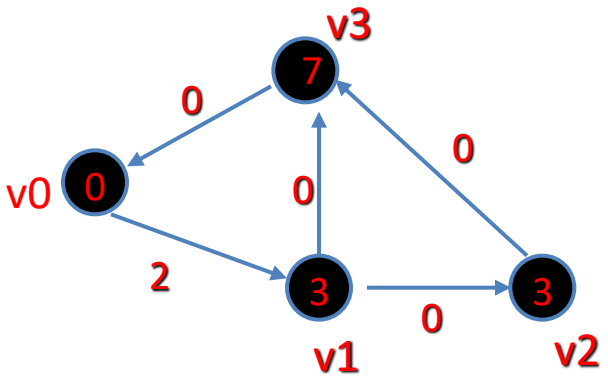
Basic Operation

In the example, *r(u) = -1, r(v) = -1* results in



- For a path $p: s \rightarrow t$, $w_r(p) = w(p) + r(t) - r(s)$
- Retardation
  - r : V→Z, an integer vertex labeling
  - $w_r(e) = w(e) + r(v) - r(u)$ for edge $e = (u,v)$
  - A retiming r is legal if $w_r(e) \geq 0, \forall e \in E$ (prove it !)

Retiming for Minimum Clock Cycle



W = register path weight matrix (minimum # latches on all paths between u and v)

D = path delay matrix (maximum delay on all paths between u and v)

W

	V0	V1	V2	V3
V0	0	2	2	2
V1	0	0	0	0
V2	0	2	0	0
V3	0	2	2	0

D

	V0	V1	V2	V3
V0	0	3	6	13
V1	13	3	6	13
V2	10	13	3	10
V3	7	10	13	7

$$c \leq \alpha \Leftrightarrow \forall p, \text{ if } d(p) > \alpha \text{ then } w(p) \geq 1$$

Conditions for legal Retiming

Assume that we are asked to check if a retiming exists for a clock cycle α

Legal retiming: $w_r(e) \geq 0$ for all e. Hence

$w_r(e) = w(e) + r(v) - r(u) \geq 0$ or

$r(u) - r(v) \leq w(e)$

For all paths p: u $\rightarrow$ v such that d(p) $\geq \alpha$, we require $w_r(p) \geq 1$

– Thus

$$1 \leq w_r(p) = \sum_{i=0}^{k-1} w_r(e_i)$$

$$= \sum_{i=0}^{k-1} [w(e_i) + r(v_{i+1}) - r(v_i)]$$

$$= w(p) + r(v_k) - r(v_0)$$

$$= w(p) + r(v) - r(u)$$

Or take the least w(p) (tightest constraint) $r(u)-r(v) \leq W(u,v)-1$

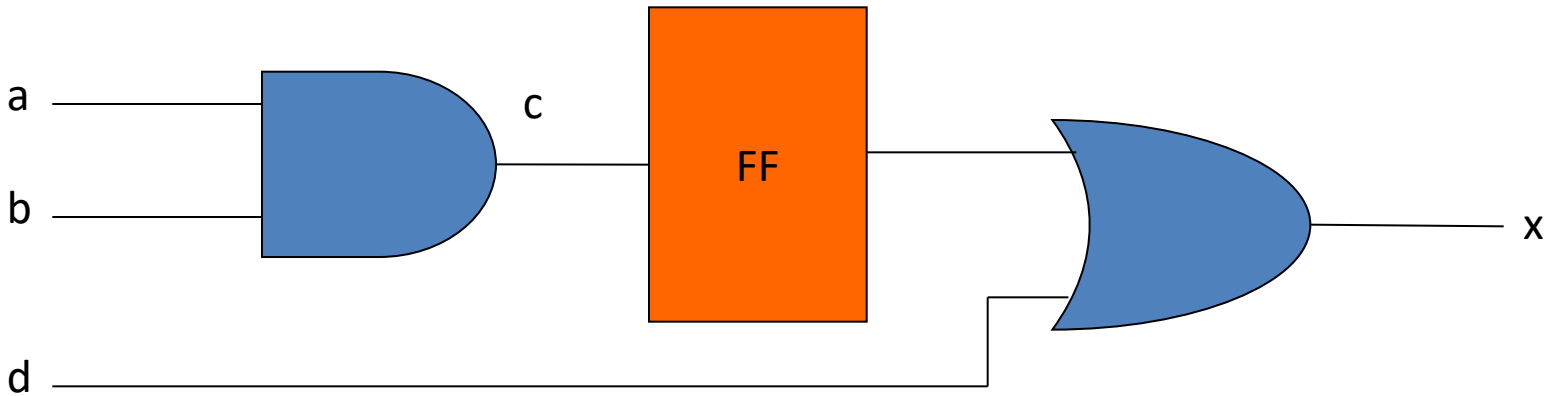
Note: this is independent of the path from u to v, so we just need to apply it to u, v such that D(u,v) > α

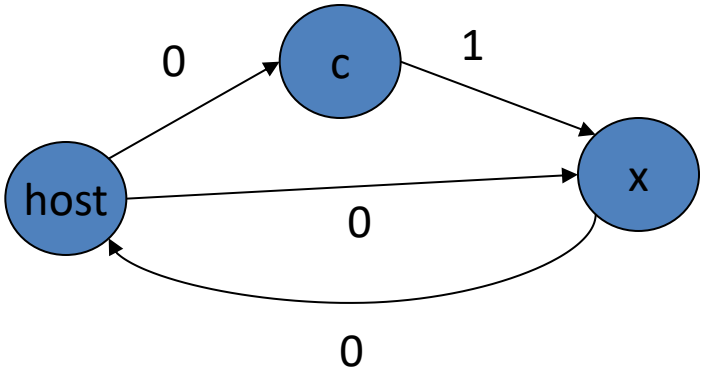
Legal Retiming

- Retiming is legal if the retimed circuit has no negative weights.
- A legally retimed circuit is functionally equivalent to the original circuit – proof by Leiserson and Saxe (1991)
- Retiming is the most general method for changing the register count and position without knowing the functions of vertices.

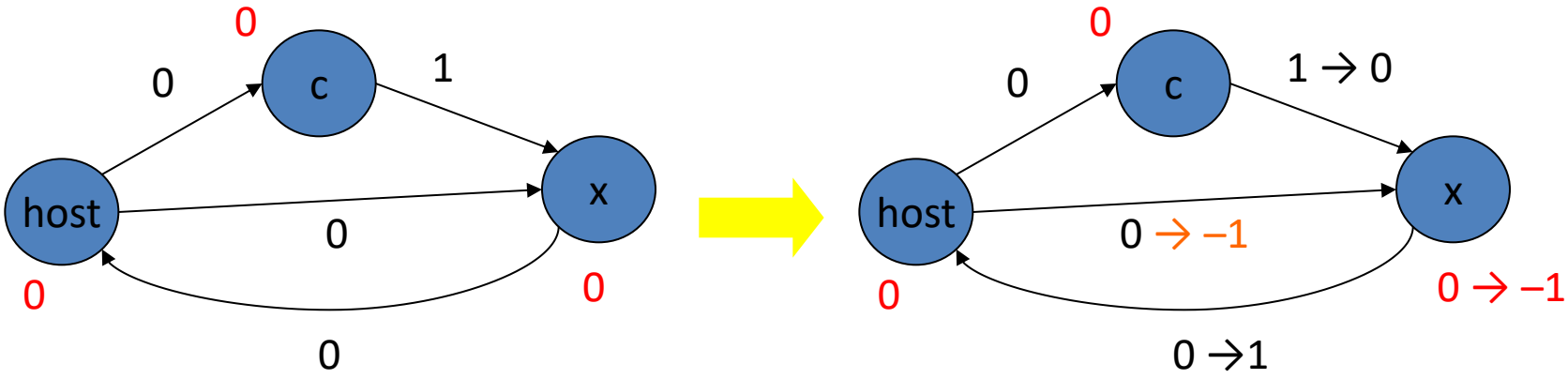
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Example



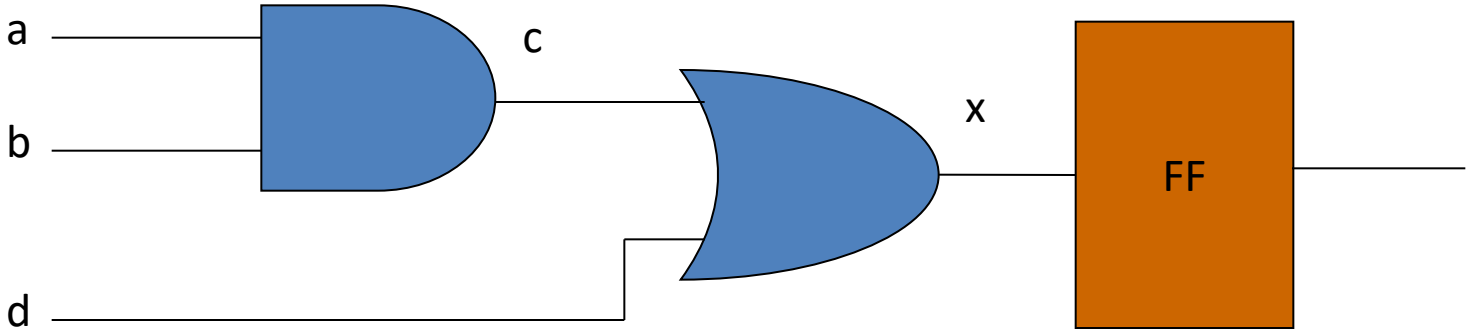


Example: Illegal Retiming

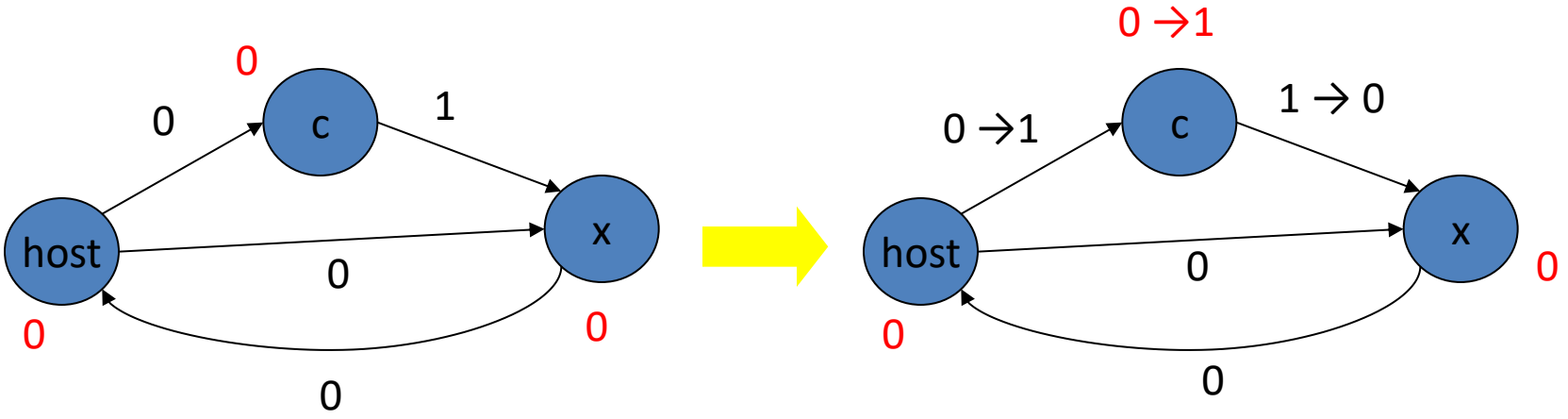


Retiming vector = {0, 0, 0}

Retiming vector = {0, 0, –1}

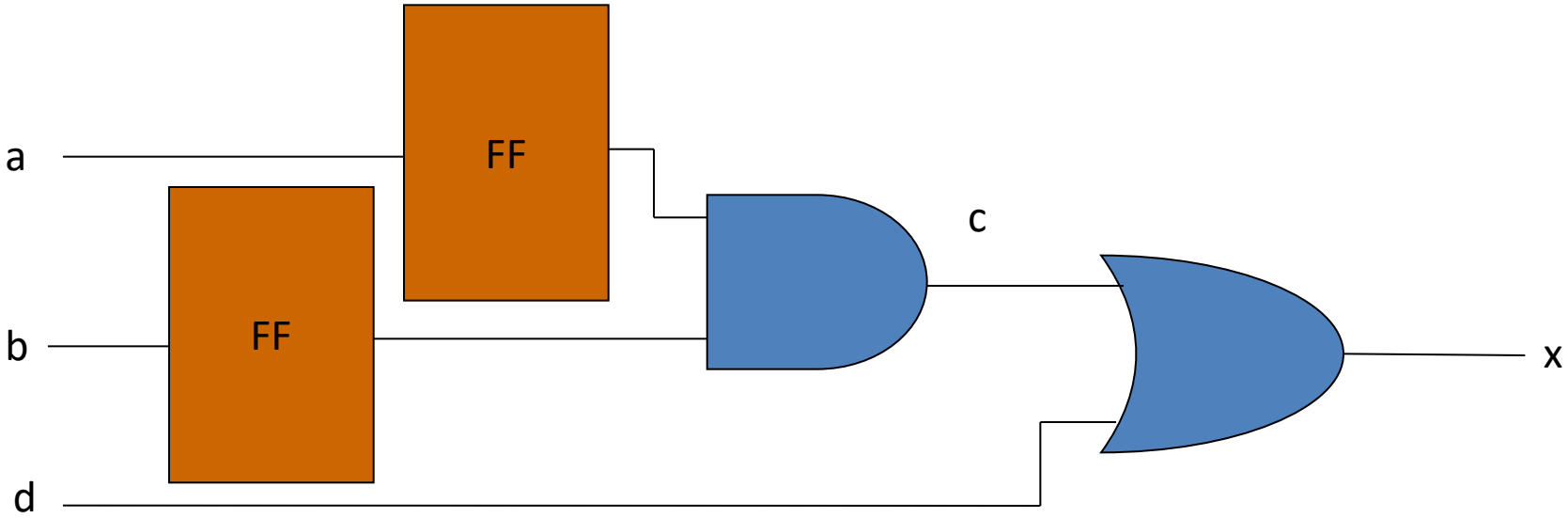


Example: Legal Retiming



Retiming vector = {0, 0, 0}

Retiming vector = {0, 1, 0}



Retiming Theorem

- Given a network G(V, E, W) and a cycle time T, (r1, . . .) is a feasible retiming if and only if:

1. $r_i - r_j \leq w_{ij}$ for all edges $(v_i, v_j) \in E$
2. $r_i - r_j \leq W(v_i, v_j) - 1$ for all node-pairs v_i, v_j such that $D(v_i, v_j) > T$

Where,

W(vi,vj): is the minimum weight for all paths between vi and vj

D(vi,vj): is the maximum delay among all minimum weight paths between vi and vj

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Proof of Condition 1

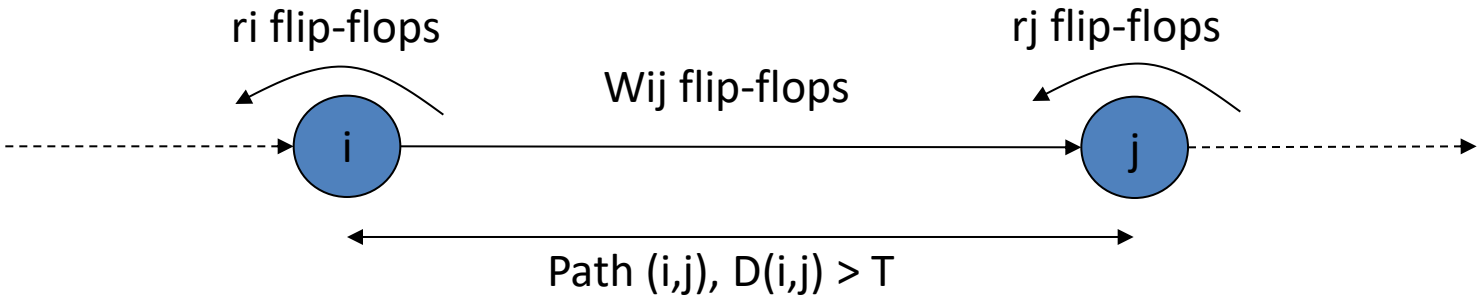
- We assume that the original network is legal, i.e., all edge weights are positive.
- For an arbitrary edge (vi,vj) ε E:

  ri – rj ≤ wij or wij + rj – ri ≥ 0, means that after retiming the new weight wij' = wij + rj – ri will be positive. Thus, condition 1 ensures the legality of retiming.

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Proof of Condition 2

- Given: d(vi) < T, for all i.
- Any retimed path whose combinational delay exceeds clock period, will have at least one flip-flop.
- The above is the requirement for correct operation.



Original weight, Wij

Retimed weight, $Wij' = Wij + rj - ri \geq 1$

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Solving the Constraints

- All constraints in difference-of-2-variable form
- Related to longest/shortest path problem

Correlator: $\alpha = 7$

Legal:
r(u)-r(v)≤w(e)

$$
\begin{aligned}
r(v_0) - r(v_1) &\le 2 \\
r(v_1) - r(v_2) &\le 0 \\
r(v_1) - r(v_3) &\le 0 \\
r(v_2) - r(v_3) &\le 0 \\
r(v_3) - r(v_0) &\le 0
\end{aligned}
$$

Timing D>7:
r(u)-r(v)≤W(u,v)-1

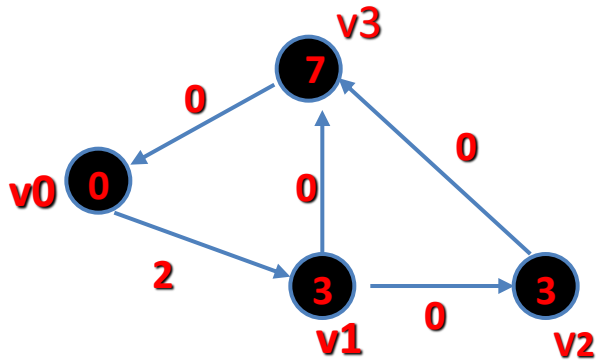
$$
\begin{aligned}
r(v_0) - r(v_3) &\le 1 \\
r(v_1) - r(v_0) &\le -1 \\
r(v_1) - r(v_3) &\le -1 \\
r(v_2) - r(v_0) &\le -1 \\
r(v_2) - r(v_1) &\le 1 \\
r(v_2) - r(v_3) &\le -1 \\
r(v_3) - r(v_1) &\le 1 \\
r(v_3) - r(v_2) &\le 1
\end{aligned}
$$

W

	V0	V1	V2	V3
V0	0	2	2	2
V1	0	0	0	0
V2	0	2	0	0
V3	0	2	2	0

D

	V0	V1	V2	V3
V0	0	3	6	13
V1	13	3	6	13
V2	10	13	3	10
V3	7	10	13	7



Solving the Constraints

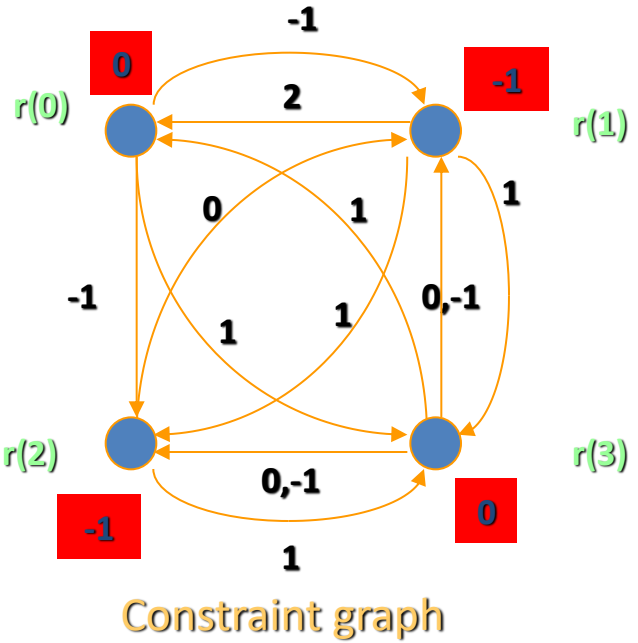
- Do shortest path on constraint graph: ($O(|V|^3)$).
- A solution exists if and only if there exists no negative weighted cycle.

Legal:
$r(u)-r(v) \le w(e)$

$$
\begin{aligned}
r(v_0) - r(v_1) &\le 2 \\
r(v_1) - r(v_2) &\le 0 \\
r(v_1) - r(v_3) &\le 0 \\
r(v_2) - r(v_3) &\le 0 \\
r(v_3) - r(v_0) &\le 0
\end{aligned}
$$

Timing D>7:
$r(u)-r(v) \le W(u,v)-1$

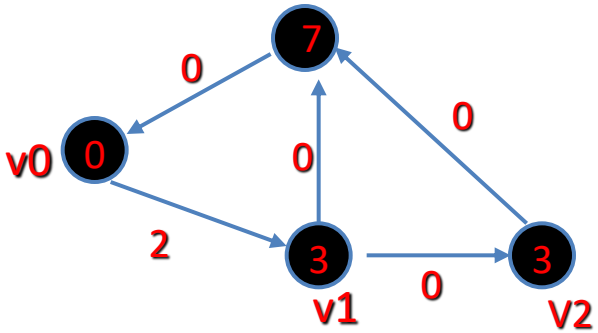
$$
\begin{aligned}
r(v_0) - r(v_3) &\le 1 \\
r(v_1) - r(v_0) &\le -1 \\
r(v_1) - r(v_3) &\le -1 \\
r(v_2) - r(v_0) &\le -1 \\
r(v_2) - r(v_1) &\le 1 \\
r(v_2) - r(v_3) &\le -1 \\
r(v_3) - r(v_1) &\le 1 \\
r(v_3) - r(v_2) &\le 1
\end{aligned}
$$



A solution: $r(v_0) = r(v_3) = 0,\ r(v_1) = r(v_2) = -1$

Retiming

To find the minimum cycle time, do a binary search among the entries of the D matrix $(O(|V|^3 \log |V|))$

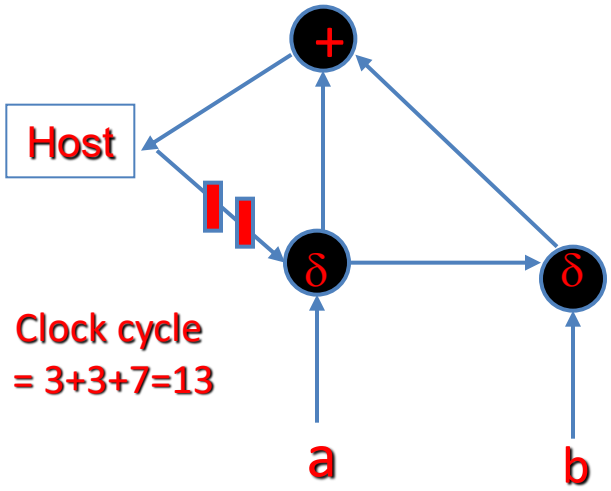


W

	V0	V1	V2	V3
V0	0	2	2	2
V1	0	0	0	0
V2	0	2	0	0
V3	0	2	2	0

D

	V0	V1	V2	V3
V0	0	3	6	13
V1	13	3	6	13
V2	10	13	3	10
V3	7	10	13	7



Retime

Retimed correlator:

